

DPCN Assignment 1

Opinion Network Formation

Person 3 — Structure, Validation & Report

Two-day build window • Graded individually

1 The two-day timeline (whole project)

Your column is **bold**. The other two are shown so you know who is waiting on you.

Block	P1 — Data & Network	P2 — Noise & Construc- tion	P3 — Structure & Report
Day 1			
Morning	Loading, encoding, missing- ness, drop rule, median fill, row-centring. Graphs 1–3. BLOCKING	Repo skeleton, <code>config.py</code> , palette, <code>requirements.txt</code> . Read MP theory.	LaTeX skeleton, Team Name + GitHub sections. Read balance theory.
Midday	HANDOFF: push centred matrix. Then Graph 9.	Graph 4: permutation null framework, 200 replicates. BLOCKING	Draft Analysis section scaffolding. Prototype Louvain on raw matrix.
Afternoon	Finish Graph 9. Wait for threshold.	Graph 5: eigendecomposition, MP band, denoised matrix.	Wait on nulls. Refine Lou- vain prototype.
Evening	Graph 8: build final network once threshold lands.	Graph 7: threshold sweep. HANDOFF: threshold + nulls.	Graph 10: communities + three-null comparison.
Day 2			
Morning	Graph 12: Deffuant–Weisbuch sweep.	Finalise Graphs 4, 5, 7 to print quality.	Graph 11: structural bal- ance across thresholds.
Midday	Write Dataset Documentation + Limitations.	Write Pipeline Followed. Build <code>run_all.py</code> .	Graph 6: reordered heatmap. Write Analysis.
Afternoon	Polish Graphs 1, 2, 3, 8, 9, 12. Write README.	Fresh-clone reproducibility check.	Write Results & Discus- sion.
Evening	Proofread. Verify own com- mits in log.	Final merge to <code>main</code> , repo pub- lic.	Assemble PDF, Individual Contribution table.

Two hard handoffs. P1 delivers the cleaned, row-centred 86×60 matrix by end of Day 1 morning. P2 delivers the denoised correlation matrix, the chosen threshold, and the null framework by end of Day 1 afternoon. If either slips, say so immediately — do not absorb it quietly, because two other people are scheduled around it.

2 Shared pipeline facts (everyone must know these)

2.1 The dataset

96 respondents $\times$ 60 Likert items, four blocks of 15: T (Technology), E (Education), S (Society/Ethics), V (Environment). Read the CSV with `encoding='utf-8-sig'` — the file carries a BOM on the first column header and will silently fail to match otherwise.

Scale maps to Strongly Disagree = -2 , Disagree = -1 , Neutral = 0 , Agree = $+1$, Strongly Agree = $+2$. **“No Comments” maps to NaN, not 0** — declining to answer is not a neutral opinion. There are 37 of them, plus 505 blanks, so 542 missing cells total.

2.2 Missingness — the decision that saves you a day

Respondents	Missing items
68	0
17	1–5
1	6–12
5	13–59
5	all 60 (empty rows)

495 of the 542 missing cells sit in 10 respondents. Drop those 10; **86 remain** with only **47 missing cells (0.9%)**. Fill with column median. Do not build a regression or Bayesian imputer — at 0.9% it cannot move any result, and the report should say you measured the problem before choosing the method.

2.3 Frozen decisions — do not change without asking the team

Filter	Drop respondents with > 12 missing → 86 remain
Imputation	Column median on the remaining 47 cells
Baseline	Row-centre: subtract each respondent's own mean across all 60 items
Similarity	Pearson correlation on the centred matrix, signs retained
Primary network	Item-item, 60 nodes. Respondent network is secondary only
Edge rule	Threshold on $ r $, chosen from the sweep, never picked arbitrarily
Null model	Column-permutation is primary; ER and rewiring are weaker comparisons

2.4 Why row-centring is mandatory

The grand mean is +1.14 on a $-2 \dots +2$ scale. Nearly everyone agrees with nearly everything. Raw-score similarity would measure who is agreeable, not what they believe.

2.5 Why the permutation null is mandatory

A correlation-derived network has transitivity built in: if A correlates with B and B with C, then A and C are pulled toward correlation automatically. That alone manufactures clustering and modularity in meaningless data. Rewiring and ER nulls take the finished graph as given and cannot detect it. The permutation null shuffles each item's column independently and reruns the *entire* pipeline, which is the only test matching the claim being made.

Evidence this matters. On the respondent network, modularity clears the rewiring null easily (0.423 vs 0.329 ± 0.011 , $z = 7.04$) but fails the permutation null (0.403 ± 0.017 , $z = 1.15$, $p = 0.08$). That community structure is a construction artefact. It is reported as a finding, not hidden.

2.6 Reference values — regression-test your code against these

- PCA on the centred item matrix: PC1–PC5 = 8.3%, 7.2%, 6.7%, 5.5%, 5.1%.
- Marchenko–Pastur: $T = 86$, $N = 60$, $Q = 1.433$, $\lambda_+ \approx 3.368$. Exactly **3** eigenvalues exceed it: 4.05, 3.98, 3.53, carrying 19.3% of variance.
- Item SD range 0.50 (E15) to 1.17 (E04). Also contested: E03 (1.12), E02 (1.12), T08 (1.09).
- Real vs shuffled item correlations: real SD = 0.136, shuffled SD = 0.108 (theory: $1/\sqrt{n-3} \approx 0.109$). Chance edge share: 66% at $|r| > 0.15$, 46% at 0.20, 33% at 0.25, 18% at 0.30.
- Block-mean correlations: T–E 0.29, T–S 0.54, T–V 0.44, E–S 0.51, E–V 0.49, **S–V 0.68**.
- Respondent network (kNN $k = 4$, **symmetrised, not mutual**): 86 nodes, 286 edges, 1 component, 7 communities sized 19/13/12/11/11/11/9. Mutual kNN fragments into 38 components — do not use it.

- Structural balance: 83% balanced triads at $|r| > 0.15$, 96% at 0.20, 100% at 0.25, against $\approx 50\%$ under sign permutation.
- Deffuant–Weisbuch on E03: 5.0 clusters for $\varepsilon \leq 0.25$, 4.2 at 0.30, 3.0 at 0.40, 1.6 at 0.50.

3 Your scope: Person 3 — Structure, Validation & Report

You own three graphs (10, 11, 6), effort-weighted to 12 points of the team’s 36. Fewest graphs, heaviest individual analyses, and the largest writing load. You also own the LaTeX build and the final PDF assembly.

You are last in the dependency chain, which means Day 1 morning is partly waiting. Use it: build the LaTeX skeleton, write the Team Name and GitHub sections, and prototype Louvain on the raw correlation matrix so that when P2’s denoised matrix and threshold land you are running within minutes rather than starting from scratch.

4 Your graphs in detail

Graph 10 — Communities vs three null models

(effort 5)

Type. Two panels. Left: the item network coloured by *detected community* (not by topic block). Right: observed modularity drawn as a vertical line against three null distributions shown as violins or histograms — Erdős–Rényi, degree-preserving rewiring, and column permutation.

The point of this figure. It is not the modularity value. It is that the answer to “is this community structure real?” depends entirely on which null you use. Expect the observed value to clear ER and rewiring comfortably while sitting close to the permutation distribution.

Method.

1. Run Louvain on the final network at P2’s operating threshold, with the shared seed. Compute modularity.
2. Pull P2’s cached permutation replicates; compute the modularity distribution across them.
3. Generate rewiring nulls with `nx.double_edge_swap` (roughly $5m$ swaps) and ER nulls with `nx.gnm_random_graph` matching n and m . 200 each.
4. Report observed value, each null’s mean $\pm$ SD, z , and p in a single table.

What to write in the caption and text. ER ignores the degree sequence entirely. Rewiring preserves degree but takes the finished graph as given, so it cannot detect structure the construction itself manufactured. Column permutation regenerates the full pipeline from scrambled responses and is therefore the only null that tests the claim actually being made. Say all three sentences — they are the report’s methodological high point.

Second analysis in this graph. Compare detected communities against the four topic labels. The interesting question is whether data-driven communities respect T/E/S/V or cut across them. Check specifically whether the **S block splits**: its questions mix technology-ethics (S03, S04, S14) with civic-inclusion themes (S07, S08, S09, S12). If it splits in the data, that is a finding about the survey instrument its designers did not intend, and it is yours.

Cut deliberately. Bootstrap community stability (resampling respondents 500 times) is the right thing to do with more time. It is not worth half a day here. Mention in Limitations that it was not done.

Graph 11 — Structural balance vs threshold

(effort 5)

Type. Line plot. X-axis: threshold. Y-axis: fraction of closed triads that are balanced. Observed line plus a shaded band for the sign-permutation null ($\approx 50\%$). Annotate the triad count at each threshold, since it falls sharply as you tighten.

The concept, in plain terms. Take any three questions that are all mutually connected. The triad is **balanced** if the pattern is internally consistent: A and B move together, B and C move

together, A and C move together. Also balanced: A and B move together, and both move opposite to C. It is **unbalanced** if A goes with B, B goes with C, but A goes *against* C — the opinion equivalent of “my friend’s friend is my enemy”. Formally, a triad is balanced when the product of its three edge signs is positive.

Method.

1. At each threshold in P2’s sweep, build the signed network: edge present if $|r| > \tau$, edge sign = $\text{sign}(r)$.
2. Enumerate all closed triads (`itertools.combinations` over nodes, checking all three edges exist — fine at 60 nodes).
3. Classify each by the product of its three signs.
4. Null: permute the edge *signs* across the existing edge set, keeping structure fixed, and recompute. 500 permutations.
5. Plot observed fraction and null band against threshold.

Expected numbers. 83% balanced at $|r| > 0.15$ (664 triads), 96% at 0.20 (137 triads), 100% at 0.25 (19 triads), against $\approx 50\%$ under sign permutation. At $|r| > 0.15$ the measured null was 0.499 ± 0.020 , giving $z \approx 16.8$.

Why the sweep matters more than any single number. At $|r| > 0.15$, two thirds of the edges are chance (see P2’s Graph 4). The balance result at that threshold alone would be weak. But the balanced fraction *rises* as the threshold tightens and noise edges are removed — 83% to 96% to 100%. That trend is the validation. Report the curve, never one value.

What it means. The class’s opinions form a coherent system rather than a contradictory jumble. There are no places where believing A implies B and B implies C but A contradicts C. This is your strongest surviving positive result, it is classic complex- networks theory (Heider balance), and very few reports at this level will contain it.

Pitfall. Triad counts collapse fast. At 0.25 you have 19 triads, which is too few to claim much on its own — which is exactly why the figure is a curve across thresholds rather than a bar at one.

Graph 6 — Reordered correlation heatmap

(effort 2)

Type. 60×60 heatmap of P2’s denoised correlation matrix, diverging colour scale centred at zero so negative correlations are visible, rows and columns reordered by your community assignment from Graph 10, with a topic-block colour strip along both margins.

Method. Order the index by community label, apply the same order to rows and columns, draw the heatmap, add the margin strips. Draw thin separator lines between communities.

What it must show. Block structure on the diagonal if communities are real. The margin strip lets a reader see immediately whether data-driven communities line up with T/E/S/V or cut across them — making your Graph 10 finding checkable by eye rather than taken on trust.

Dependency. Needs P2’s matrix *and* your own community output, so it is last. That is fine — it is the cheapest of your three.

Pitfall. Use a diverging colourmap centred at zero, not a sequential one. With a sequential map, a correlation of -0.4 and one of $+0.05$ can look similar, which destroys the whole point of retaining signs.

5 Your report sections

You own the majority of the writing. Budget Day 2 afternoon and evening properly.

- **Team Name** and **GitHub link** — trivial, but they are required sections in a required order. Do them Day 1 morning while waiting.
- **Analysis and Visualizations** (shared with P2’s Pipeline section). Every figure must be referenced in the text. A figure nobody discusses reads as padding. Make each visualization answer

a stated question: how connected is the network, what groups emerge, is the structure robust, is it stronger than chance.

- **Results and Discussion.** Distinguish descriptive facts from interpretation. A good sentence pattern: “At threshold τ , the GCC contains X of 60 questions. The detected communities contain questions from [blocks]. This indicates [interpretation], subject to the chosen similarity measure and threshold.”
- **Individual Contribution.** Build it from CONTRIBUTIONS.md, not from memory. Include commit hashes. The assignment is graded individually and this table is the evidence.

5.1 The headline to write toward

The class appears to agree on almost everything, but that consensus is superficial. Once each respondent’s agreement baseline is removed, exactly three statistically real opinion dimensions survive Marchenko–Pastur filtering (19.3% of variance). The opinion space is internally coherent: 83–100% of closed triads are balanced against $\approx 50\%$ by chance. Topic domains couple unevenly — Ethics and Environment travel together ($r = 0.68$) while Technology and Education are nearly independent ($r = 0.29$). And the apparent student factions do not survive a properly constructed null, so we report them as a construction artefact rather than a finding.

6 Your LaTeX and assembly duties

- Day 1 morning: `report/report.tex` skeleton with the seven required sections in the required order, plus a figure-inclusion macro so everyone drops files in consistently.
- Use `\includegraphics` on the .pdf versions of figures, not the PNGs — vector stays sharp and examiners often read on screen.
- Day 2 evening: compile, check the page count is close to 8, check every figure is referenced in the body text, check no figure is illegible at print size.
- Final check before submission: the GitHub link in the report actually resolves, and the repo is public.

7 GitHub: structure and workflow

7.1 Repository layout

```

opinion-network/
  CLAUDE.md           <- project context, read first
  README.md           <- setup + how to reproduce everything
  requirements.txt     <- pinned versions
  run_all.py           <- regenerates every figure from scratch
  CONTRIBUTIONS.md    <- append-only task log
  data/
    Survey_Results_UC.csv <- READ ONLY, never edit in place
  src/
    common/
      config.py        <- SEED, paths, colour palette (import, never
        redefine)
      loader.py         <- P1: parse, encode, filter, impute, centre
      nulls.py          <- P2: permutation / rewiring / ER generators
    person1/ graphs_1_2_3.py graph_9.py graph_8.py graph_12.py
    person2/ graph_4.py graph_5.py graph_7.py
    person3/ graph_10.py graph_11.py graph_6.py
  figures/
    person1/ person2/ person3/
  notebooks/          <- exploration only, nothing load-bearing
  report/
    report.tex

```

`figures/``<- final copies used in the PDF`

7.2 Rules that keep the repo clean

- Modules are importable and side-effect free. No plotting or file writes at import time — `run_all.py` orchestrates everything.
- Everyone imports `src/common/config.py` for SEED, paths, and the colour palette. Never hard-code a seed inline.
- No absolute paths. Resolve relative to repo root with `pathlib`.
- Figures write to `figures/<owner>/` at ≥ 150 DPI, then get copied into `report/figures/` only once finalised.

7.3 Branching and commits

- Work on `person1-work` / `person2-work` / `person3-work`. Never commit directly to `main`. Open a PR when a graph is finished.
- Commit message format: `[P<n>] [G<graph#>] what changed`. Example: `[P2] [G5] Add Marchenko-Pastur eigenvalue filtering`.
- One commit per meaningful step, **not** one giant commit per graph. The assignment is graded individually and `git log` is the evidence — a single squashed commit at the end looks like one person did all the work.
- Never commit on a teammate's behalf and never squash multiple people's work together.

7.4 CONTRIBUTIONS.md

Append one line per completed task, as you finish it:

```
2026-09-17 | P2 | Graph 5: MP denoising + eigenvalue figure | a3f91c2
```

This file becomes the literal source for the report's Individual Contribution section. Do not try to reconstruct it from memory on Day 2 evening.

7.5 Before submission

Repo public. README has working clone-and-run instructions. `python run_all.py` succeeds from a completely fresh clone in an empty directory — verify this yourself rather than assuming it works because it works on your machine.

8 How to create the graphs (standards for all 12)

8.1 Shared visual conventions

Agree the hex codes before anyone plots, then put them in `config.py` and import them. Every figure in the report must use the same palette for T/E/S/V or the reader cannot compare across figures.

- One qualitative 4-colour palette for topic blocks, used in every figure that distinguishes them.
- A separate diverging palette for signed/correlation heatmaps, centred at zero so negative correlations are visibly distinct from weak positive ones.
- Matplotlib only. No seaborn-specific styling that teammates cannot reproduce.
- White background, no grid unless it aids reading, no chartjunk.

8.2 Every figure must have

1. A descriptive title (or none, if the LaTeX caption carries it — pick one convention team-wide and stick to it).
2. Axis labels with units where relevant.
3. A legend whenever more than one series is drawn.
4. Legibility *at the size it appears in the PDF*. Test by compiling `report.tex` and looking at the page, not by viewing the PNG full-screen. Font sizes that look fine at 100% zoom routinely become unreadable at half-column width.

8.3 Saving

```
fig.savefig(FIGDIR / "graph_N_name.png", dpi=200, bbox_inches="tight")
fig.savefig(FIGDIR / "graph_N_name.pdf", bbox_inches="tight") # vector, for LaTeX
```

Prefer the PDF version in `report.tex` — vector figures stay sharp at any zoom and examiners often read on screen.

8.4 Reproducibility

Every stochastic call takes the shared seed: Louvain, rewiring, permutation nulls, spring layouts, Deffuant–Weisbuch. An unseeded spring layout alone will make your figure differ every run and your teammates’ reproduction will not match yours.

9 Things nobody on the team should do

- Do not claim polarisation. PC1 explains 8.3% — the data does not support a two-camp story.
- Do not report modularity, clustering, or any structure metric without its null comparison beside it in the same table or figure.
- Do not use rewiring or ER as primary evidence.
- Do not use mutual kNN for the respondent network.
- Do not build an imputation model beyond column median.
- Do not report a single threshold without the sensitivity sweep.
- Do not present the Deffuant–Weisbuch low- ϵ plateau without noting the discrete-Likert artefact.
- Do not silently change a frozen decision. Ask the team.

The report’s headline. The class appears to agree on almost everything, but that consensus is superficial. Beneath it there are exactly three statistically real opinion dimensions, the opinion space is internally coherent (83–100% balanced triads vs $\approx 50\%$ by chance), and topic domains couple unevenly (Ethics–Environment $r = 0.68$, Technology–Education $r = 0.29$). The apparent student factions are a construction artefact, and we say so.